

Male reproductive organs are at risk from environmental hazards

Male reproductive disorders that are of interest from an environmental point of view include sexual dysfunction, infertility, cryptorchidism, hypospadias and testicular cancer. Several reports suggest declining sperm counts and increase of these reproductive disorders in some areas during some time periods past 50 years. Except for testicular cancer this evidence is circumstantial and needs cautious interpretation. However, the male germ line is one of the most sensitive tissues to the damaging effects of ionizing radiation, radiant heat and a number of known toxicants. So far occupational hazards are the best documented risk factors for impaired male reproductive function and include physical exposures (radiant heat, ionizing radiation, high frequency electromagnetic radiation), chemical exposures (some solvents as carbon disulfide and ethylene glycol ethers, some pesticides as dibromochloropropane, ethylendibromide and DDT/DDE, some heavy metals as inorganic lead and mercury) and work processes such as metal welding. Improved working conditions in affluent countries have dramatically decreased known hazardous workplace exposures, but millions of workers in less affluent countries are at risk from reproductive toxicants. New data show that environmental low-level exposure to biopersistent pollutants in the diet may pose a risk to people in all parts of the world. For other toxicants the evidence is only suggestive and further evaluation is needed before conclusions can be drawn. Whether compounds as phthalates, bisphenol A and boron that are present in a large number of industrial and consumer products entails a risk remains to be established. The same applies to psychosocial stressors and use of mobile phones. Finally, there are data indicating a particular vulnerability of the fetal testis to toxicants—for instance maternal tobacco smoking. Time has come where male reproductive toxicity should be addressed from entirely new angles including exposures very early in life.